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MLDL

Experiment No 01

Aim: Implement Linear and Logistic Regression on real-world datasets

1. Dataset Source

Two different real-world datasets were used in this experiment to implement Linear Regression and Logistic Regression.

Linear Regression Dataset

- **Dataset Name:** Advertising Dataset
- **Source:** Kaggle
- **Link:** https://www.kaggle.com/datasets/ashydv/advertising-dataset

Logistic Regression Dataset

- **Dataset Name:** Breast Cancer Wisconsin (Diagnostic) Dataset
- **Source:** Kaggle (UCI ML Repository)
- **Link:** https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

Each algorithm uses a different dataset, as required.

2. Dataset Description

2.1 Advertising Dataset (Linear Regression)

This dataset is used to predict product sales based on advertising expenditure.

- **Number of instances:** 200
- **Number of features:** 4

Features:

- **TV:** Advertising budget spent on TV
- **Radio:** Advertising budget spent on radio
- **Newspaper:** Advertising budget spent on newspapers
- **Sales:** Product sales (target variable)

Target Variable:

- Sales (continuous numerical value)

Dataset Characteristics:

- Fully numerical dataset
- No missing values
- Suitable for regression problems

2.2 Breast Cancer Dataset (Logistic Regression)

This dataset is used to classify tumors as benign or malignant.

- **Number of instances:** 569
- **Number of features:** 30 numerical features

Target Variable:

- **diagnosis**
    - 0 → Benign
    - 1 → Malignant

Dataset Characteristics:

- Medical classification dataset
- Binary output
- Clean and well-structured

3. Mathematical Formulation of the Algorithm

3.1 Linear Regression

Linear Regression models the relationship between independent variables and a dependent variable using a linear equation.

$$y=\beta 0+\beta 1x+\varepsilon$$

Where:

- y = predicted value
- x = input feature
- β0 = intercept
- β1 = slope
- ϵ = error term

The model minimizes **Mean Squared Error (MSE)**:

$$\boxed{MSE = \frac{1}{n}\sum(y_i - \hat{y}_i)^2}$$

3.2 Logistic Regression

Logistic Regression is used for binary classification problems and uses the sigmoid function.

$$\boxed{\sigma(z) = \frac{1}{1 + e^{-z}}}$$

The model is optimized using **log loss (cross-entropy loss)**:

$$\boxed{Loss = -\frac{1}{n}\sum[y\log(p) + (1 - y)\log(1 - p)]}$$

4. Algorithm Limitations

Linear Regression Limitations

- Assumes linear relationship between variables
- Sensitive to outliers
- Cannot handle non-linear patterns

Logistic Regression Limitations

- Assumes linear decision boundary
- Sensitive to outliers
- Not suitable for complex non-linear classification

5. Methodology / Workflow

5.1 Overall Workflow of the Experiment

The overall workflow followed in this experiment is shown below:

1. Selection of suitable real-world datasets from Kaggle

2. Loading the datasets into the Python environment
3. Understanding dataset structure and features
4. Data preprocessing (removal of unnecessary columns and encoding labels)
5. Selection of input features and target variable
6. Splitting the dataset into training and testing sets
7. Training the machine learning models
8. Making predictions using the trained models
9. Evaluating model performance using appropriate metrics
10. Visualization of results for better interpretation

5.2 Workflow for Linear Regression

The workflow for Linear Regression using the Advertising dataset is as follows:

1. Load the Advertising dataset
2. Select advertising budget (TV) as the independent variable
3. Select Sales as the dependent (target) variable
4. Visualize the relationship between TV budget and Sales using a scatter plot
5. Split the dataset into training and testing sets
6. Train the Linear Regression model on training data
7. Predict sales values for test data
8. Evaluate the model using Mean Squared Error (MSE) and R^2 score
9. Plot the regression line to visualize model performance

5.3 Workflow for Logistic Regression

The workflow for Logistic Regression using the Breast Cancer dataset is as follows:

1. Load the Breast Cancer dataset
2. Remove unnecessary columns (if present)
3. Encode diagnosis labels into binary values (0 and 1)
4. Select medical features as input variables
5. Select diagnosis as the target variable
6. Split the dataset into training and testing sets
7. Train the Logistic Regression model
8. Predict class labels and probabilities
9. Evaluate the model using accuracy and confusion matrix
10. Plot ROC curve to analyze classification performance

6. Performance Analysis

Linear Regression Performance

- **Mean Squared Error (MSE):** Measures prediction error
- **R^2 Score:** Indicates goodness of fit

Interpretation:
Higher R^2 value shows that advertising expenditure has a strong influence on sales.

Logistic Regression Performance

- **Accuracy:** Measures classification correctness
- **Confusion Matrix:** Shows correct and incorrect predictions
- **ROC Curve & AUC:** Measures classifier performance

Interpretation:
The model effectively classifies tumors as benign or malignant with high accuracy

7. Hyperparameter Tuning

Linear Regression

- Linear Regression has no major hyperparameters
- Performance depends on data quality and feature selection

Logistic Regression

Hyperparameters tuned:

- **max_iter:** Increased for better convergence
- **Regularization strength (C):** Controls overfitting

Impact of Tuning:

- Improved convergence
- Better model stability
- Higher accuracy

CODE:
Linear Regression – Advertising Dataset

```
import pandas as pd
import matplotlib.pyplot as plt
```

```python
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load data
data = pd.read_csv("advertising.csv")
print(data.head())

X = data[['TV']]
y = data['Sales']

plt.scatter(X, y)
plt.xlabel("TV Advertising")
plt.ylabel("Sales")
plt.title("TV vs Sales")
plt.show()

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("MSE:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

plt.scatter(X_test, y_test)
plt.plot(X_test, y_pred)
plt.xlabel("TV Advertising")
plt.ylabel("Sales")
plt.title("Linear Regression Output")
plt.show()
```

Logistic Regression – Breast Cancer Dataset

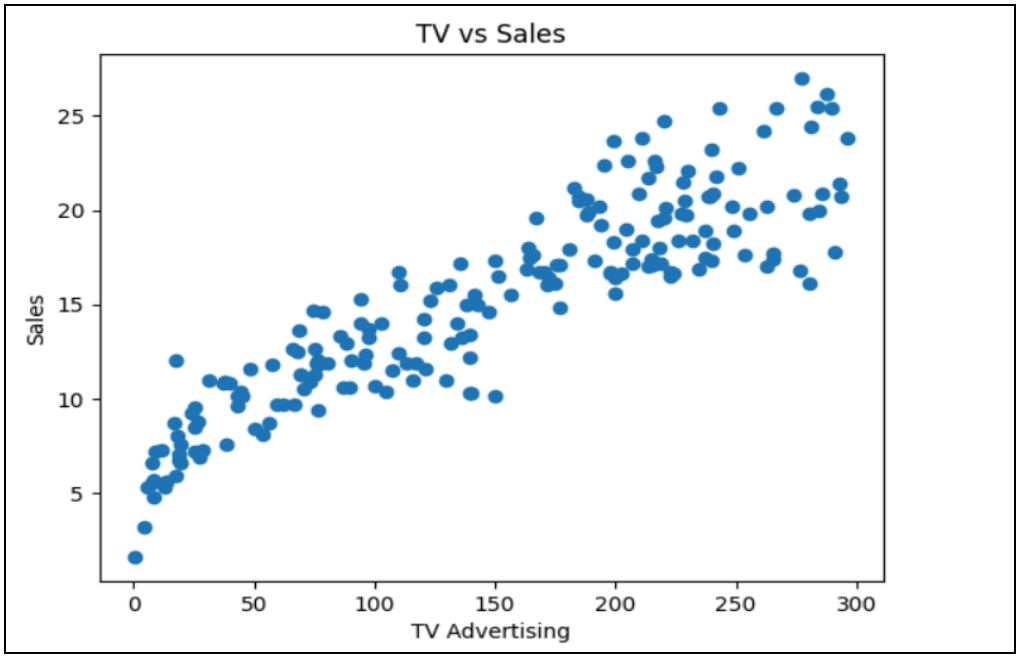
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, roc_curve,
auc

# Load dataset
data = pd.read_csv("breast-cancer.csv")
print(data.head())
print(data.columns)

# Drop columns ONLY if they exist
data = data.drop(columns=['id', 'Unnamed: 32'], errors='ignore')

# Encode target column
data['diagnosis'] = data['diagnosis'].map({'M': 1, 'B': 0})

# Features and target
X = data.drop('diagnosis', axis=1)
y = data['diagnosis']

# Class distribution
plt.hist(y)
plt.xlabel("Class (0 = Benign, 1 = Malignant)")
plt.ylabel("Count")
plt.title("Breast Cancer Class Distribution")
plt.show()

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = LogisticRegression(max_iter=5000)
model.fit(X_train, y_train)
```

```python
# Predictions
y_pred = model.predict(X_test)
y_prob = model.predict_proba(X_test)[:, 1]

print("Accuracy:", accuracy_score(y_test, y_pred))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
plt.imshow(cm)
plt.title("Confusion Matrix")
plt.colorbar()
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# ROC Curve
fpr, tpr, _ = roc_curve(y_test, y_prob)
roc_auc = auc(fpr, tpr)

plt.plot(fpr, tpr, label="AUC = %.2f" % roc_auc)
plt.plot([0, 1], [0, 1])
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()
```

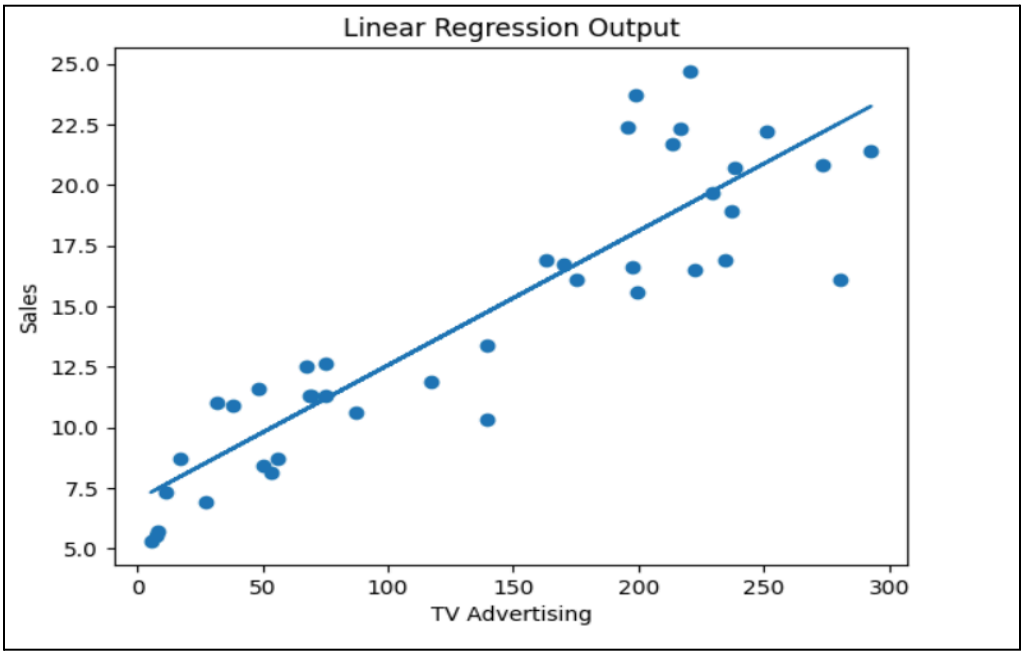
OUTPUT:

Linear Regression :

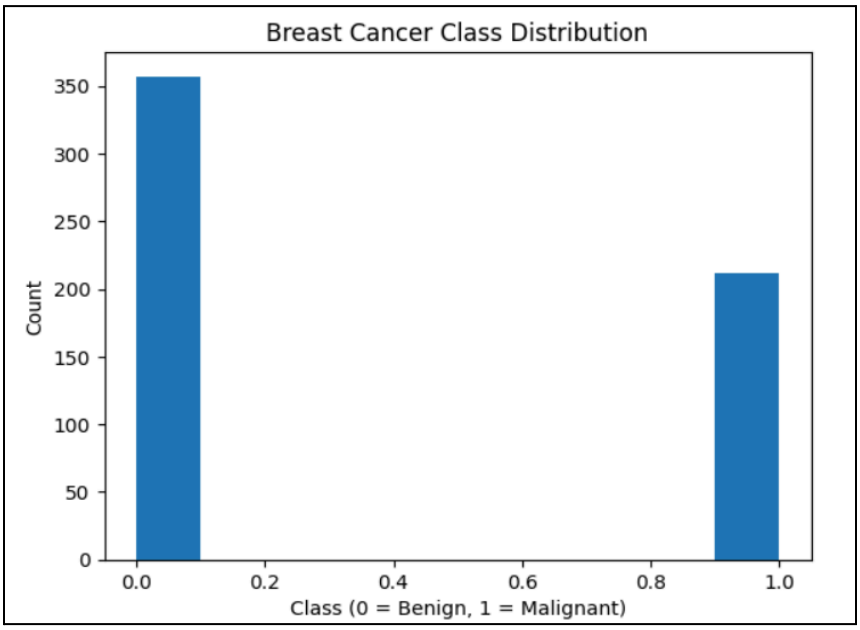
	TV	Radio	Newspaper	Sales
0	230.1	37.8	69.2	22.1
1	44.5	39.3	45.1	10.4
2	17.2	45.9	69.3	12.0
3	151.5	41.3	58.5	16.5
4	180.8	10.8	58.4	17.9

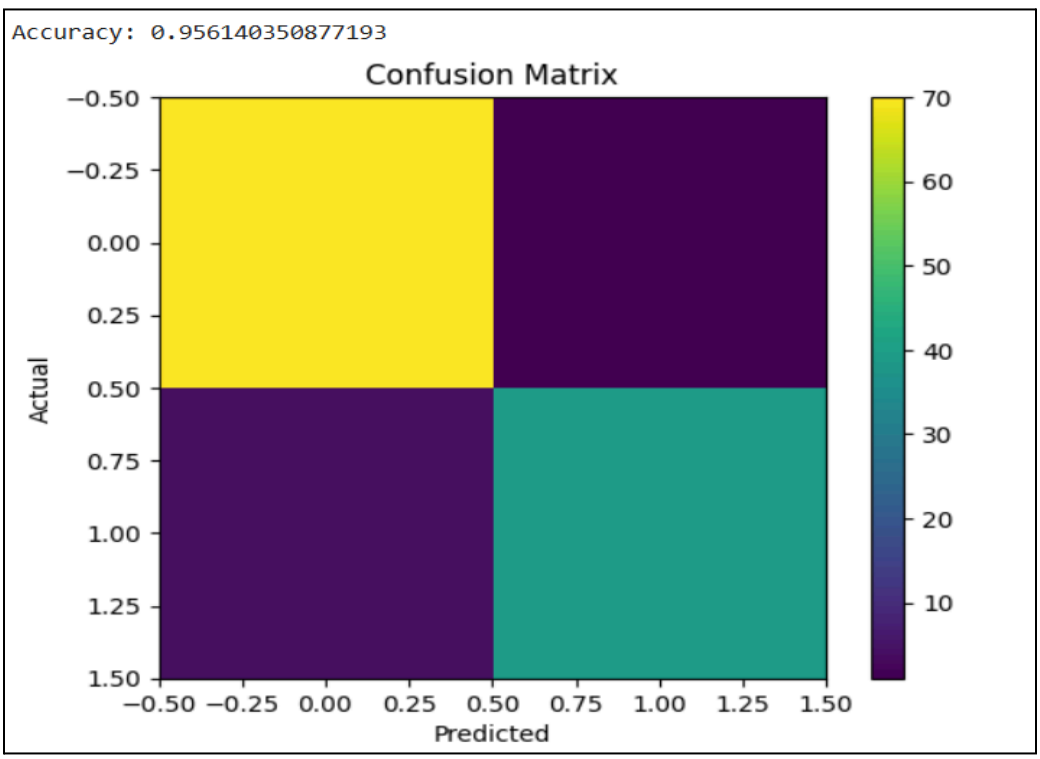


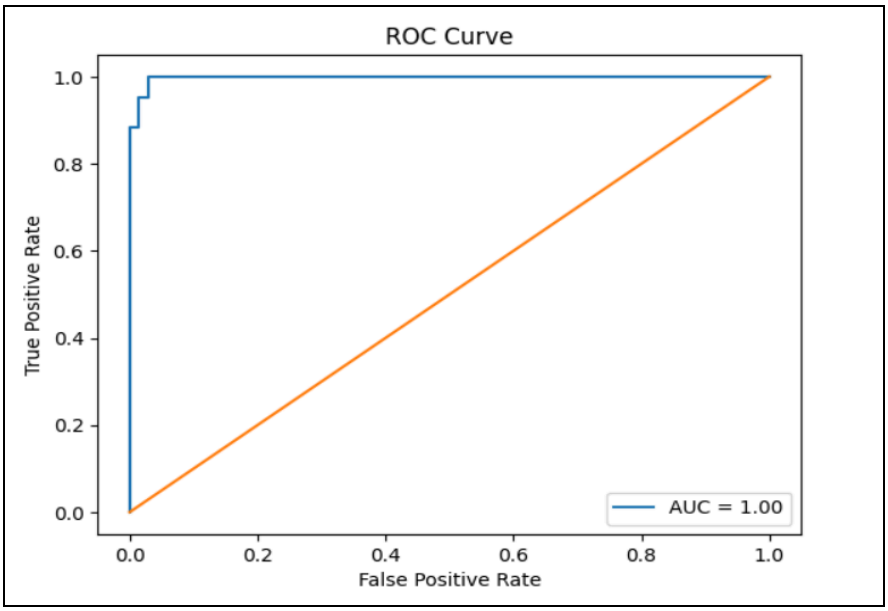

```
MSE: 6.101072906773963
R2 Score: 0.802561303423698
```



Logistic Regression :







Conclusion:

In this experiment, Linear Regression and Logistic Regression were successfully implemented on two different real-world datasets. Linear Regression was used to predict continuous sales values, while Logistic Regression was used to classify medical data into binary classes. The models were evaluated using appropriate metrics and visualizations, demonstrating effective learning and prediction capabilities.